

Research on Fair, Accountable, and Transparent ML systems

(Explaining what learned models predict: In which cases can we trust machine learning models and when is caution required)

Introduction

Our lives are becoming more dependent on the presence of machine learning(ML). But as data, the fuel of ML, is collection of pervasive human bias and disparate factors in society, ML is also vulnerable to bias that affects the fairness [1]. Since ProPublica reported COMPAS, a tool used by courts in the US, was biased against African-Americans [2], similar studies have been conducted in other fields, such as YouTube's favoritism to specific dialect and gender [3], or an A-level grading system that discriminated the marginalized students [4], which left us a question 'can we still trust ML models?'.

This essay will address

- 1) Impossibility of bias-free data and ML.
- 2) Conflict in concepts of individual and group fairness.
- 3) How explainability can lead ML models to trust.

keywords: machine learning, bias, fairness, explainability, accountability, regulation

Why can't we just trust ML models?

1. No data nor ML algorithms can be bias free.

As Tom Mitchell puts, ML models learn the patterns from historical data and generalize them to predict unseen data [5]. Unfortunately, Human intervention is inevitable in the ML pipeline [6]. Consequently, our subconscious biases are fed into algorithms and the algorithms will make the predictions as humans designed them, reflecting the human biases or even amplifying them throughout the ML pipeline [7][8][9].

It is difficult to pick out what is biased and what is not biased in our data because biases are subconsciously ingrained in our society. Amazon's recruiting artificial intelligence(AI) system, for example, combed through applicants' résumés from the past 10 years, where preference on men over women was prevalent, resulting in discrimination against women [10]. The applicants' qualifications were not AI's concern, its only concern was the patterns.

Even if we eliminate the features that we think cause discrimination in our data, there will normally be proxies and other correlated features that will eventually reveal the features inadvertently, leading our models to undesired and unfair results [11]. Eliminating biases completely is not feasible, hence neither are biases in models. Thus, we should take great heed to the sources of the biases and mitigate their negative impact on the prediction [12].

2. There is no single definition of fairness in machine learning

As difficult as it is to define bias with one definition, fairness is also difficult to define with a single definition in machine learning [13][14]. In attempt to define algorithmic fairness, many researchers have tried to bring the legal concept of fairness into machine learning; disparate treatment and disparate impact [7].

One simple notion of fairness is to achieve the same proportion of a protected class from groups with different demographic values, using the concept of statistical parity [15]. This concept, however, is not as fair as it sounds at a distance. Given that protected class is 'Asian/Non-Asian' – an Asian student

may excel in his/her academic achievement, while he/she belongs to a minority group. But he/she may be classified into subnormal just to satisfy the statistical parity constraint. As a result, despite one's own qualifications, he/she will only be classified into either as a true positive or a false negative.

This brings up the major dilemma in the algorithmic fairness – groups versus individuals. The statistical parity approach will suffice to achieve the fairness of the majority because ML algorithms are more affected by the majority [16], only to accelerate the polarization of the society. On the other hand, if we attempt to achieve the fairness of individuals [17], we may protect the marginalized classes, but we may end up resulting in overfitting, sacrificing the greater good(the overall accuracy) [18]. It is clear not only that interpretation of the fairness is context dependent but also that there is no single approach to satisfy both individual and group fairness.

What are the requirements for trustworthy ML?

1. Prerequisite to trust: Explainability

We simply cannot just trust the ML models by themselves. On the bottom line is the nature of ML models being unable to explain the causality [19]. To overcome this obstacle, we need qualitative approaches in addition to quantitative approaches; explainability.

Although such metrics as trustworthiness cannot be computed, experts' prior knowledge in the application domain can help humans whether to accept or to reject a prediction if ML models can explain their reasoning [20]. This is how explainability builds trust upon the ML models.

Furthermore, another great benefit we can leverage from explainable ML models is its strong ability to visualize the behavior of ML models, facilitating debugging and audit [21][22]. For example, Y. Ahn [23] shows how visualization can help humans understand the behavior of a ML model, identify the latent variables(causality), and mitigate inherent biases, hence getting proximate to fairness. These benefits of explainability enables ML models to evoke related essential concepts to trust AI, such as transparency and accountability, which therefore makes the root basis of trustworthy ML [24].

2. Fundamentals to trust: Accountability

But who is to blame if a decision made by ML leads to negative social impact or serious business loss? Putting the people involved in the ML pipeline, such as ML practitioners or decision makers, back in the frame will act as a drive for stakeholders to trust ML models, should ML based decisions go wrong; the notion on which this is based is 'accountability' [25].

Since the concepts(explainability, fairness, accountability, transparency) are still abstract to be measured within society, many countries and organizations including EU-GDPR [26] are focusing on establishing regulations in attempt to avoid adverse impact [22]. These concepts are intertwined with one another, under human oversight [27] if required, make up fundamentals for trustworthy AI.

Conclusion

Behind the predictions are interconnected biases that are omnipresent in human society, which in turn emerge on our ML models. Therefore, it is humans' task to infer and alleviate the causation of disparate impact that such biases may bring into society. In this context, explainable AI plays an important role, which not only enables humans to scrutinize the inner behavior of ML models but also provides fundamentals to trust ML models. However, the extent to which individuals find them fair varies and is heavily dependent on the context in which ML models are deployed. Thus, this is our next challenge to define clear and fair regulations on which all members of society can agree to achieve trustworthy ML models.

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